CHEMISTRY II
FSPC 0034

PAST YEAR TEST 1, SEMESTER 3, 2019

(a)

Q.1

- **Open systems** – both matter (mass) and energy **(0.5)** can move into and out of an open system **(0.5)**. The earth, for example, is an open system. Energy comes in from the sun, and is re-radiated back to space. **(0.5** or any related example)
- **Closed systems** – matter can't be transferred from a closed system, but energy can **(0.5).** Example is the "coffee cup" calorimeter used in many general chemistry laboratories. These are great ways to see how a system (some chemical reaction) causes a change in temperature in the surroundings (the solvent, often water) that is contained in the cup. (**0.5** or any related example)
- **Isolated systems** – neither matter or energy (**0.5**) can be transferred from these systems to the surroundings (**0.5).** Example is, soup in an insulated container **(0.5** or any related example)

(b)

- (i)

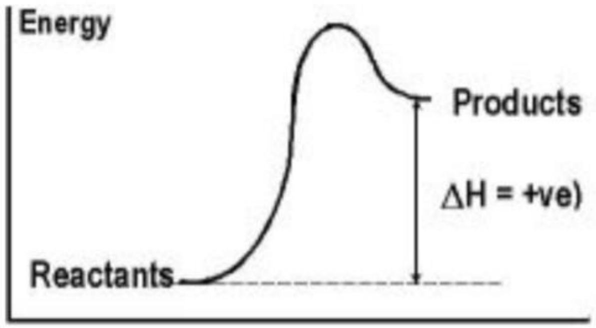
q = m x Cs x ΔT

= (4.0 g) x (0.385 J/g.0C) x (47.0 0C)

= 72.4 J

- (ii)

Energy profile diagram of an endothermic reaction



Reaction profile

Q.1

(b)

Q.1

(ii)

- Correct process: endothermic **(0.5)**
- Correct x axis: reaction profile/coordinates **(0.5)**
- Correct y axis: energy **(0.5)**
- Correct label: product higher than reactants **(0.5)**
- Correct label ΔH: + 72.4 J **(1)**

(c)

Q.1

(i) Enthalpy of Combustion

$= [m(H_2O) \times 4.184\ J\ g^{-1}\ °C^{-1} \times \Delta T] + [C\ (comb) \times \Delta T]$

$= 3.5\times10^2 g \times 4.184\ J\ g^{-1}\ °C^{-1} \times 1.5°C) + (1769\ J\ °C^{-1} \times 1.5°C)$

$= 4.9\times10^3\ J$

0.2g magnesium released 4.9×10^3 J heat

24.31g/mol magnesium released

$= (4.9\times10^3\ J \times 24.31 g\ mol^{-1}) \div 0.2g$

$= 5.89\times10^5\ J\ mol^{-1}$ or $5.89\times10^2\ kJ\ mol^{-1}$

Thus, enthalpy of combustion of magnesium is $-5.89\times10^2\ kJ\ mol^{-1}$

(ii) $Mg(s) + \frac{1}{2} O_2(g) \rightarrow MgO(s)$ $\Delta H_c = -5.89\times10^2$ kJ/mol

(c)

Q.1

(iii) The enthalpy changes when **one mole (0.5)** of a pure compound is **completely burned in excess oxygen (0.5)** under standard conditions

Q.1

(d)

$$2NO(g) + 3/2\ O_2(g) \rightarrow N_2O_5(s) \qquad \Delta H = -223.7\ kJ$$

$$2(NO_2(g) \rightarrow NO(g) + 1/2\ O_2(g)) \quad \Delta H = 2(57.1\ kJ)$$

$$2NO(g) + 3/2\ O_2(g) \rightarrow N_2O_5(s) \qquad \Delta H = -223.7\ kJ$$

$$2NO_2(g) \rightarrow 2NO(g) + O_2(g) \qquad \Delta H = 114.2\ kJ$$

$$\cancel{2NO(g)} + 1/2.\cancel{3/2}\ O_2(g) + 2NO_2(g) \rightarrow N_2O_5(s) + \cancel{2NO(g) + O_2(g)}$$

$$2NO_2(g) + 1/2\ O_2(g) \rightarrow N_2O_5(s) \qquad \Delta H = \text{-}109.5\ kJ$$

Q.2

(a)

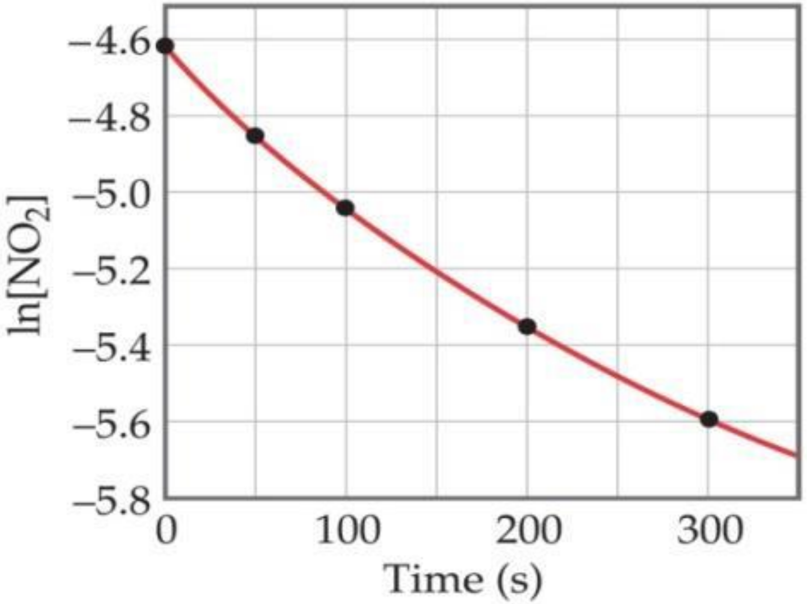
(i)

Time (s)	$[NO_2]$; M	ln $[NO_2]$	$1/[NO_2]$; M^{-1}
0.0	0.01000	-4.610	100
50.0	0.00787	-4.845	127
100.0	0.00649	-5.038	154
200.0	0.00481	-5.337	208
300.0	0.00380	-5.573	263

(a)

Q.2

A graph of ln / [NO_2] vs t :



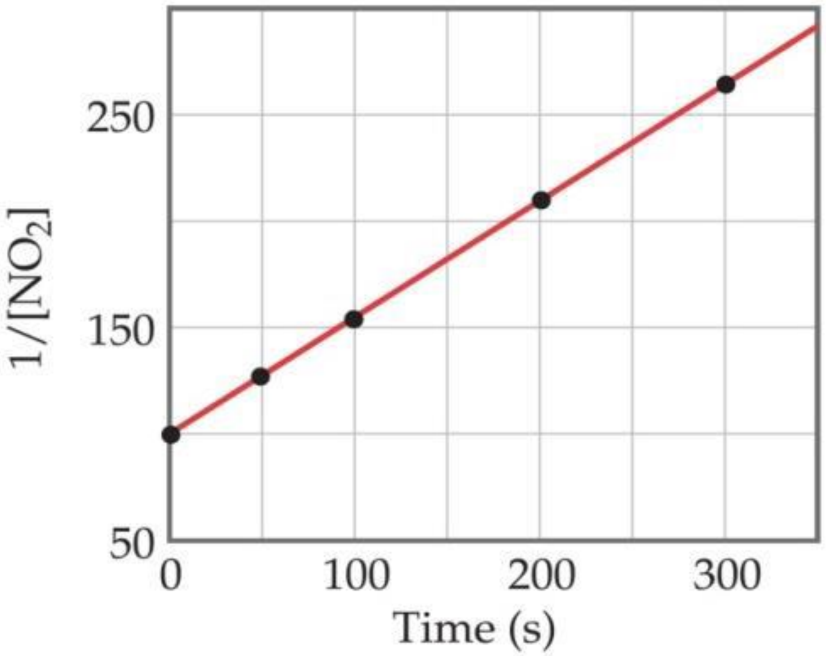
$$\ln [A]_t = -kt + \ln [A]_0$$

The plot is not a straight line, so the process is not first-order in [NO2]. Does not fit

(a)

Q.2

A graph of $1 / [NO_2]$ vs t :



$$1 / [A]_t = kt + 1 / [A]_0$$

This is a straight line. Therefore, the process is second-order in [NO2].

Q.2

(a)

(ii) : *Rate* $= k\,[NO_2]^2$

(iii) : *Appropriate calculations* : $k = 0.545\ M^{--1}\ s^{-1}$

(iv) : *Half life* $t_{1/2}$

$= 1/(k[A]_0)$

$= 183.5\ s$

(b)

Q.2

- Order of reaction = First order → Rate = k[N_2O_5]

- For a first order reaction :
- $t_{1/2}$ = 0.693 / k

  = 0.693 / 2.56 × 10^{-4} s^{-1}

  =2 707 s

(Note: Students can also convert 2 707 s into minute. In this case, the answer is 45.1 minutes)

Q.2

(b)

$$\text{Reaction rate} = k[N_2O_5] = (2.56 \times 10^{-4}\ s^{-1})\ (3.5 \times 10^{-3}\ mol\ dm^{-3})$$

$$\therefore Rate = 8.96 \times 10^{-7} mol\ dm^{-3} s^{-1}$$

$$k = \frac{2.30}{t} \log \frac{initial\ concentration}{concentration\ at\ time\ T}$$

$$2.56 \times 10^{-4} s^{-1} = \frac{2.303}{200\ minutes \times \frac{60\ seconds}{minute}}\ log \frac{initial\ concentration}{3.1 \times 10^{-2}\ mol\ dm^{-3}}$$

$$\therefore Initial\ concentration = 0.67\ mol\ dm^{-3}$$

(c).(i)

Compare Exp.1 and 2 since [Y] and [Z] remains constant

$$\left(\frac{0.2}{0.1}\right)^x = \left(\frac{2}{1}\right)$$

$$2^x = 2$$

$$x = 1$$

∴ Order of reaction with respect to X is first order.

Compare Exp.1 and 3 since [X] and [Z] remains constant

$$\left(\frac{0.2}{0.1}\right)^x = \left(\frac{4}{1}\right)$$

$$2^x = 4$$

$$x = 2$$

∴ Order of reaction with respect to Y is second order.

Q.2

(c).(i)

Q.2

Compare Exp.1 and 4 since [X] and [Y] remains constant

$$\left(\frac{0.2}{0.1}\right)^x = \left(\frac{1}{1}\right)$$

$$2^x = 1$$

$$x = 0$$

∴ Order of reaction with respect to Z is zero order.

Q.2

(c)

(ii) : Rate = $k[X][Y]^2$

(iii)

Use data in Exp.1, and substitute into rate equation :

$1\ mol/dm^3.s^1 = k\ [0.1\ mol/dm^3]\ [0.1\ mol/dm^3]^2$

$\therefore\ k = 1000\ mol^{-2}.dm^6.s^{-1}$

(a)

Q.3

(i) Calculate the Qc : $Qc = (NH_3)^2 / (N_2).(H_2)^3$

$= (0.406)^2 / (0.0785).(0.0960)^3$

$= 2.37 \times 10^3$

Since *Qc* is not equal to *K*c, it is not at equilibrium.

Q.3

(a)

(ii) Since Qc bigger than Kc, the net reaction will proceed from right to left. NH_3 will decrease and increasing N_2 and H_2 until Qc = Kc = 1.7×10^2.

(b)

Q.3

- Calcium in seawater will react with carbonate producing calcium carbonate. This in turn will reduce the carbonate concentration.

- The equilibrium will proceed towards producing more carbonate (to the right), thus forcing more dissolution of carbon dioxide from the air, reducing earth temperature.

Q.3

(c)

$$
\begin{aligned}
K\text{sp} &= (Ca^{2+})^3(PO_4^{\;3-})^2 \\
&= (2.01 \times 10^{-8})^3\,(1.6 \times 10^{-5})^2 \\
&= 2.08 \times 10^{-33}
\end{aligned}
$$

(d)

Q.3

Use ICE Method : Write equilibrium constant expression

Kc = $[HI]^2[H_2][I_2]$ = 54.3 , then

Obtain Qc : Qc = $(0)^2(0.5)(0.5)$ = 0

Determine the direction based on comparing Qc and Kc :

Qc < Kc , so the reaction shift to the right to achieve equilibrium.

(d)

Reaction Equation	$H_2(g)$	+ $I_2(g)$	→ $2HI(g)$
Initial Concentration	0.5	0.5	0
Change	-x	-x	+2x
Equilibrium	0.5-x	0.5-x	2x

Determine value x

$$K_c = \frac{(2x)^2}{(0.5-x)(0.5-x)} = 54.3$$

$$\frac{(2x)^2}{(0.5-x)^2} = 54.3$$

$$\frac{2x}{0.5-x} = \sqrt{54.3} = 7.368$$

x = 0.393 M

Determine the concentration

$[H_2] = (0.500 - 0.393)\ M = 0.107\ M$

$[I_2] = (0.500 - 0.393)\ M = 0.107\ M$

$[HI] = (2 \times 0.393)\ M = 0.786\ M$

Remember…
You Must Pass!